Date: 19th December 2025 **Project supervisor:** Prof. MICHAIL XYNTARAKIS
Course: Data Analysis & Visualization

Global Football Talent Flows: Network and Visual Analysis Using FIFA Data (2017–2022)

Author: Tripti Singh, Nikita Sharma Gyawali

Research Questions, Data Sources, and Analytical Approach

Research Questions

This project aims to analyze the **structure, distribution, and interconnectedness of professional football players** across clubs, leagues, and nationalities using a consolidated FIFA player dataset covering the years **2017 to 2022**. Rather than focusing on match outcomes or game-level performance, the project emphasizes **structural and relational patterns** within professional football.

The analysis is guided by the following research questions:

1. How are football players distributed across clubs, leagues, and nationalities from 2017 to 2022?

This question investigates how players are allocated across different clubs and leagues and how national representation varies within these structures over multiple years. By examining the distribution of players across organizational levels, the study seeks to understand broader patterns of talent concentration, diversity, and representation in professional football.

2. Which clubs and leagues act as central hubs in the football talent network?

This question focuses on identifying clubs and leagues that play a **central role** in the global football ecosystem. By modeling the dataset as a network, the analysis highlights entities that are highly connected to players from diverse national backgrounds. These hubs often reflect strong recruitment power, international reach, and structural importance within football's global hierarchy.

3. How does the club–league–nation structure change over time between 2017 and 2022?

By combining multiple years of data, this question explores **temporal changes** in player distribution, club composition, and league internationalization. This longitudinal perspective allows the study to observe whether certain leagues become more globally diverse over time, whether clubs increase or decrease their international reach, and how structural relationships evolve across seasons.

Collectively, these research questions are designed to uncover **relational patterns and structural dynamics** rather than on-field performance metrics, making them particularly well suited for **network analysis and exploratory visual analytics**.

Data Sources

FIFA Player Dataset (2017–2022)
The analysis is based on a **cleaned FIFA player dataset obtained from Kaggle**, originally compiled from official FIFA player records.

- **Source:** Kaggle
  FIFA 23 Official Dataset – Clean Data
  https://www.kaggle.com/datasets/kevwesophia/fifa23-official-datasetclean-data
- **Years used:** 2017–2022
- **File used:** FIFA_2017_2022_COMBINED.xlsx
- **Type:** Structured football player data

The Kaggle source provides player-level datasets for multiple years. For this project, datasets from **2017 to 2022** were combined into a single consolidated dataset by **appending rows and adding a year identifier**, creating a longitudinal dataset suitable for temporal and network-based analysis.

The **2023 dataset was intentionally excluded** because several key columns were missing or inconsistent relative to earlier years. Including it would have reduced comparability across years and potentially compromised data integrity. Excluding 2023 ensured that all analyses were conducted on a **consistent and structurally aligned dataset**.

Data Dictionary:

Column Name	Description	Type
ID	Unique numeric identifier assigned to each player in the FIFA dataset.	Integer
Name	Full name of the player as listed in official FIFA records.	Text
Age	Player's age during the FIFA season.	Integer
Photo	URL or path to the player's profile image.	Text
Nationality	Player's country of nationality.	Text
Flag	URL or path to the national flag image representing the player's nationality.	Text
Overall	Overall rating representing the player's current ability level.	Integer
Potential	Estimated maximum rating the player can achieve in future seasons.	Integer
Club	Club team the player is contracted to for the given season.	Text
Club Logo	URL or path to the club's official logo image.	Text
Value(£)	Estimated market value of the player in British pounds.	Numeric
Wage(£)	Weekly wage of the player in British pounds.	Numeric
Special	Composite score derived from the sum of key player attributes.	Integer
Preferred Foot	Player's dominant foot used for passing and shooting (Left or Right).	Text
International Reputation	Reputation score reflecting global recognition (scale of 1–5).	Integer
Weak Foot	Rating of the player's ability to use their non-dominant foot (1–5 scale).	Integer
Skill Moves	Rating of the player's dribbling and technical flair (1–5 scale).	Integer
Work Rate	Player's attacking and defensive work rate (e.g., High/Medium).	Text
Body Type	Player's physical build classification used by the FIFA engine.	Text
Real Face	Indicates whether the player has an authentic scanned face in the game (Yes/No).	Boolean
Position	Primary on-field position played by the player (e.g., ST, CM, CB).	Text
Jersey Number	Squad number worn by the player at their club.	Integer
Joined	Date the player joined their current club.	Date
Loaned From	Parent club if the player is currently on loan.	Text
Contract Valid Until	Year when the player's current club contract expires.	Integer
Height(cm.)	Player's height measured in centimeters.	Integer
Weight(lbs.)	Player's weight measured in pounds.	Integer
Crossing	Ability to deliver accurate crosses into the penalty area.	Integer
Finishing	Ability to score goals in close-range situations.	Integer
HeadingAccuracy	Accuracy and effectiveness when heading the ball.	Integer
ShortPassing	Accuracy and speed of short-range passes.	Integer

Column Name	Description	Type
Volleys	Ability to strike the ball cleanly while airborne.	Integer
Dribbling	Skill in controlling the ball while moving.	Integer
Curve	Ability to bend the ball when passing or shooting.	Integer
FKAccuracy	Accuracy when taking free kicks.	Integer
LongPassing	Accuracy and vision for long-distance passes.	Integer
BallControl	Ability to control the ball under pressure.	Integer
Acceleration	Speed at which the player reaches top pace.	Integer
SprintSpeed	Maximum running speed of the player.	Integer
Agility	Ability to move and change direction quickly.	Integer
Reactions	Speed of response to in-game situations.	Integer
Balance	Ability to maintain stability when challenged.	Integer
ShotPower	Power behind shots taken by the player.	Integer
Jumping	Vertical leap ability, important for aerial duels.	Integer
Stamina	Ability to maintain performance throughout the match.	Integer
Strength	Physical strength when contesting possession.	Integer
LongShots	Ability to score goals from long distances.	Integer
Aggression	Intensity and competitiveness in physical play.	Integer
Interceptions	Ability to anticipate and intercept opponent passes.	Integer
Positioning	Ability to position effectively when attacking.	Integer
Vision	Awareness and creativity in spotting passes.	Integer
Penalties	Ability to take penalty kicks successfully.	Integer
Composure	Ability to remain calm under pressure.	Integer
Marking	Ability to track and mark opposing players defensively.	Integer
StandingTackle	Effectiveness of standing tackles.	Integer
SlidingTackle	Effectiveness of sliding tackles.	Integer
GKDiving	Goalkeeper's diving ability.	Integer
GKHandling	Goalkeeper's ball-handling skill.	Integer
GKKicking	Goalkeeper's kicking and distribution ability.	Integer
GKPositioning	Goalkeeper's ability to position correctly.	Integer
GKReflexes	Goalkeeper's reaction speed to shots.	Integer
Best Position	Computed optimal playing position based on attributes.	Text
Best Overall Rating	Overall rating corresponding to the player's best position.	Integer
Year_Joined	Calendar year the player joined their current club.	Integer
fifa_Year	FIFA edition year (e.g., 2017, 2018, … 2022).	Integer

Dataset Structure and Content

Each row in the combined dataset represents **one player in one year**, and includes:

- Player identifiers and demographic information (name, age, nationality)
- Club and league affiliation

- Player ratings (overall, potential)
- Technical and physical attributes (e.g., passing, shooting, dribbling)
- Economic attributes (player value and wage)

This structure enables both **cross-sectional analysis** (within a given year) and **longitudinal analysis** (across years).

Data Suitability

The dataset is particularly well suited for:

- **Network analysis** of club–league–nation relationships
- **Longitudinal analysis** across multiple seasons
- **Visualization of structural patterns** in professional football
- Exploratory analysis of similarity, clustering, and outliers

Analytical Approach and Visualization Plan

Interactive and High-Quality Visualizations

A core emphasis of this project is the creation of **interactive, high-quality visualizations**, inspired by prior coursework (HW3 Part 1 and HW4 Part 1). Visualizations are designed not only to present data, but to **communicate analytical insights clearly and effectively**.

Using tools such as **Tableau** and **network visualization software (e.g., JavaScript)**, the project includes:

- Interactive dashboards with filters for **year, league, club, and nationality**
- Scatter plots, distributions, and summary charts that allow users to explore player attributes dynamically
- Clear visual encodings (size, color, position) to emphasize comparisons and patterns

Careful attention is paid to layout, labeling, annotation, and filtering to ensure that visualizations remain readable and analytically meaningful, consistent with best practices demonstrated in earlier coursework.

Advanced Analytics: Similarity, Dimension Reduction, and Outliers

To move beyond basic descriptive statistics, the project incorporates **advanced analytical techniques** that support exploratory and comparative analysis:

Multidimensional Scaling (MDS)

MDS is used to project high-dimensional player attribute data into a two-dimensional space. Players with similar technical and physical profiles appear closer together, making similarity relationships visually interpretable and enabling the identification of clusters and patterns.

Outlier Detection

Outlier detection methods, such as distance-based approaches or **Local Outlier Factor (LOF)**, are applied to identify players whose attribute profiles differ significantly from the majority. These outliers often represent unique playing styles, extreme strengths, or unusual attribute combinations that warrant closer examination.

Together, these methods support **exploratory data analysis** and help uncover patterns that are not immediately visible in raw tabular data.

Graph Analytics and Network Visualizations

A central component of the project is **graph-based analysis**, where the football dataset is modeled as a network rather than a flat table.

Key network structures include:

- **Player–Club relationships**
- **Club–League relationships**
- **Player–Nationality relationships**

These relationships are combined into a **multi-layer (club–league–nation) network**, enabling analysis of:

- Which clubs and leagues act as **high-connectivity hubs**
- How national representation varies across leagues
- How structural relationships evolve over time (2017–2022)

Network analytics such as **degree centrality, community detection, and visual clustering** are used to interpret these structures. Force-directed layouts and clustering techniques help reveal dominant hubs, tightly connected communities, and peripheral entities within the network.

Research Inspiration and Prior Work

The analytical approach is informed by:

- Prior coursework emphasizing **story-driven visualization**
- Published sports analytics research that applies **dimension reduction and clustering** to compare players
- Network-based studies examining **talent flow and organizational structure** in professional sports

Rather than replicating existing analyses, this project adapts these ideas to the FIFA dataset, integrating **visual analytics, statistical techniques, and network modeling** into a cohesive exploratory framework.

Summary

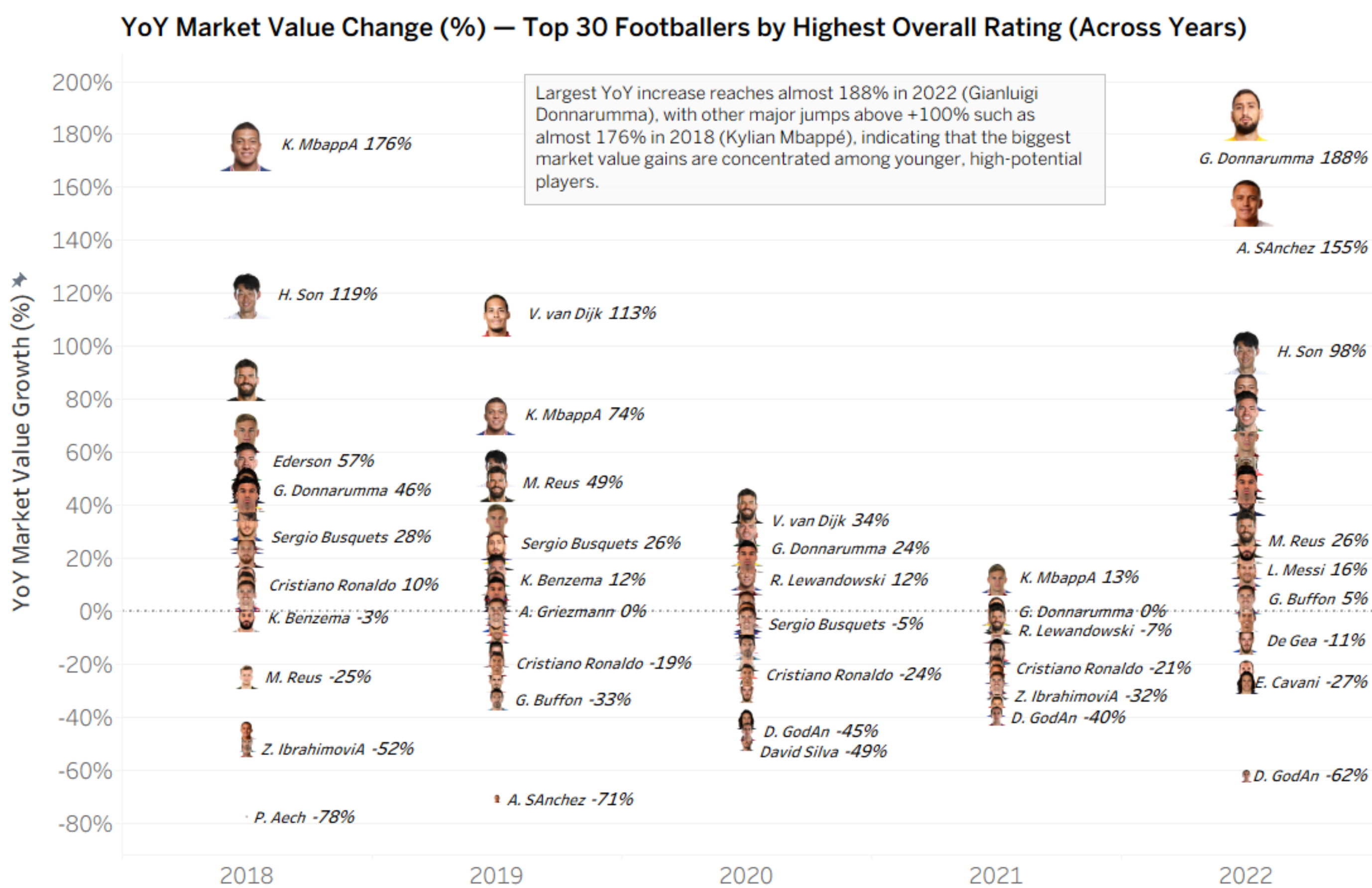
By integrating:

- Interactive, high-quality visualizations
- Advanced analytics (MDS & outlier detection)
- Graph and network analysis

this project provides a **multi-perspective exploration** of football player data from **2017 to 2022**. The approach aligns closely with course objectives and demonstrates how complex datasets can be effectively analyzed through **both visual and analytical lenses**, yielding insights into the structure and dynamics of professional football.

YoY Market Value Change (%) of Top 30 Footballers (FIFA)

This chart presents the **year-over-year (YoY) percentage change in market value** for the **Top 30 footballers by overall rating** from **2018 to 2022**.



YoY market value change ranges roughly from −78% to +188% (2018–2022), with extreme gains (>100%) concentrated among emerging stars and sharp declines (<−30%) more common among veterans.

Extreme positive growth (breakout seasons)

- **Kylian Mbappé** recorded **+176% (2018)** and **+74% (2019)** YoY growth, marking the strongest early-career revaluation.
- **Gianluigi Donnarumma** peaked at **+188% (2022)**, the **highest YoY increase** observed in the dataset.
- **Heung-Min Son** achieved **+119% (2018)** and **+98% (2022)**, indicating sustained market appreciation across multiple seasons.

YoY gains above 100% reflect rapid market reassessment, typically associated with age, breakout performance, and long-term contract value.

Moderate growth among elite but established players

- **Virgil van Dijk** shows **+113% (2019)** followed by more stable growth of **~+34% (2020)**.
- **Joshua Kimmich** records consistent appreciation, ranging from **+34% to +68%** across seasons.
- **Alisson** maintains positive YoY growth between **+31% and +88%**, indicating steady valuation without extreme volatility.

These players demonstrate market stability rather than speculative spikes.

Negative YoY changes concentrated among aging players

- **Zlatan Ibrahimović** declines by **−52% (2018)** and **−32% (2021)**.
- **Diego Godín** records **−62% (2022)**, one of the steepest declines.
- **Cristiano Ronaldo** shows repeated depreciation: **−19% (2019)**, **−24% (2020)**, **−21% (2021)**.

Negative YoY values reflect aging curves, contract length, and reduced resale potential rather than immediate performance.

Increasing dispersion after 2020

- Post-2020, YoY values range from **+188% to −62%**, compared to a tighter range in earlier seasons.
- This widening spread indicates **greater valuation volatility**, with young players accelerating and veterans depreciating faster.

Central tendency around zero

- Many players cluster between **−10% and +15% YoY**, especially in **2020–2021**, suggesting short-term market corrections and stabilization phases.

takeaway

Despite similar **overall ratings**, YoY market value changes vary widely:

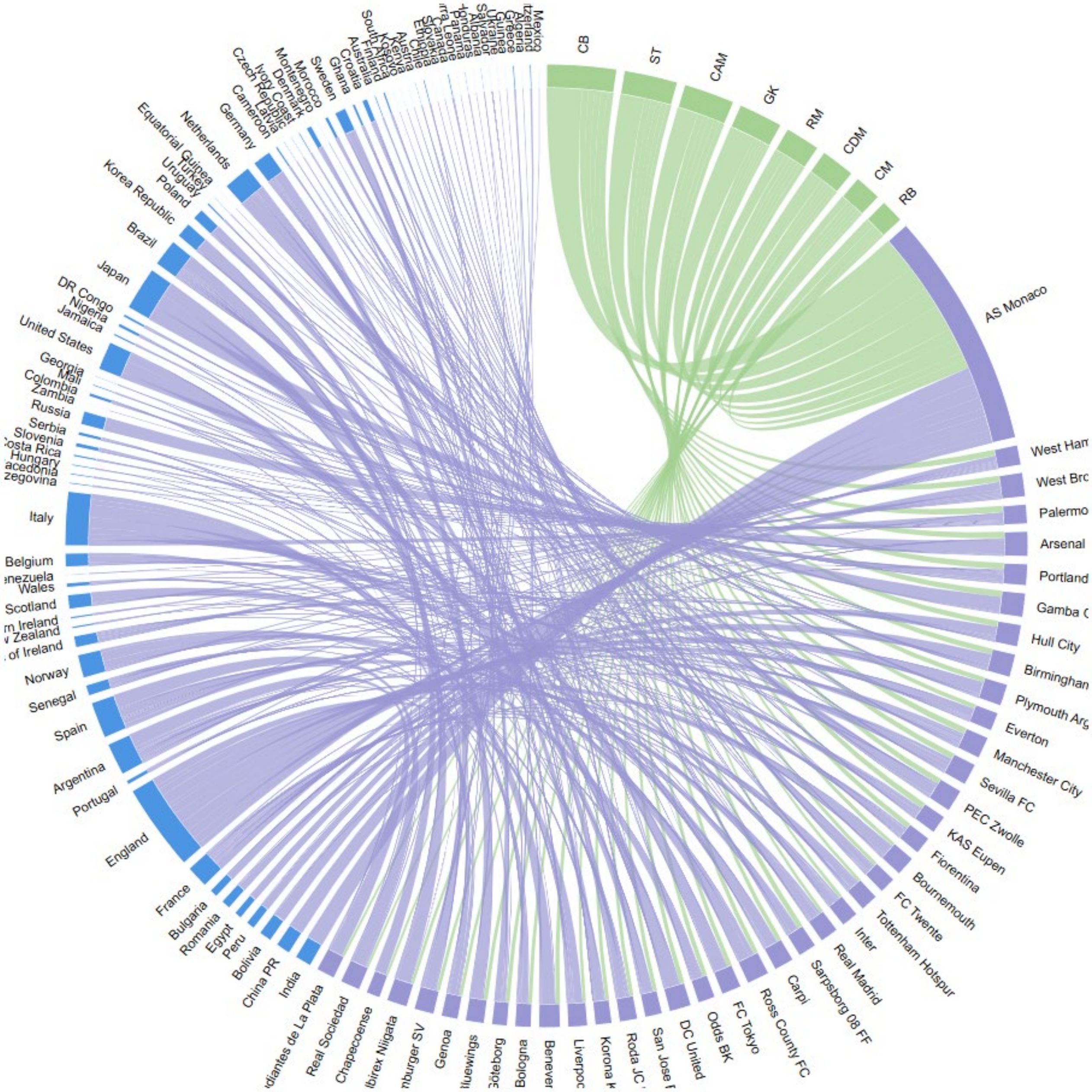
- **Range:** −78% to +188%
- **Breakout threshold:** > +50% YoY
- **Decline threshold:** < −30% YoY

This confirms that **market value growth is driven more by age, future potential, and timing than by static skill level alone**.

Overall, the year-over-year market value trends highlight that elite footballers do not experience uniform valuation trajectories, even when selected at the same overall rating level. Market value growth is strongly influenced by age, career stage, and future potential rather than current ability alone, as evidenced by the explosive gains among younger players such as Mbappé and Donnarumma contrasted with consistent declines among aging veterans like Ronaldo and Ibrahimović. The widening dispersion of YoY changes in later seasons further suggests increasing market volatility and sharper differentiation between emerging talent and established stars, reinforcing that transfer market valuations are forward-looking and reward long-term potential over short-term performance.

Interactive Chord Network: Nationality -> Club -> Best Position (Season-wise) – Created in JavaScript

This chord diagram visualizes the **global talent pipeline in football**, linking **player nationalities (left)** to **playing positions (top)** and ultimately to **clubs (right)**. The width of each arc represents the **volume of players** flowing through that pathway, allowing us to identify concentration, diversity, and recruitment strategies.



1. Nationality patterns (left side)

- A **small number of football-dominant countries** (e.g., **France, England, Spain, Argentina, Brazil, Portugal**) generate **disproportionately large flows**.
- These countries act as **primary talent suppliers**, feeding players across multiple positions and clubs.
- Many other countries appear with **thin but numerous connections**, indicating **broad global scouting** rather than reliance on a few regions.

2. Position as a sorting mechanism (top)

- Positions such as **ST, CAM, CM, RM, and CB** show the **thickest inbound flows**, meaning most international talent is concentrated in:
    - **Attacking roles** (ST, CAM, RM)
    - **Central control roles** (CM, CDM, CB)

- Goalkeepers (GK) and fullbacks (RB) have **fewer but more specialized pathways**, reflecting lower turnover and more selective recruitment.

3. Club-level structure (right side)

- Clubs differ strongly in how they connect to nationalities and positions:
  - **Elite destination clubs** (e.g., Real Madrid, Manchester City, Inter, Arsenal) show **many inbound links across positions**, reflecting squad depth and financial capacity.
  - **Mid-tier or development clubs** show **broader nationality diversity but narrower positional focus**.

4. AS Monaco's distinctive role

- **AS Monaco stands out as a high-connectivity hub** despite its smaller league and market size.
- Monaco receives:
  - **Players from many nationalities**
  - **Primarily attacking and midfield profiles**
- This pattern indicates Monaco's role as a **development and stepping-stone club**, specializing in:
  - Identifying young, high-potential talent
  - Developing them before transfer to elite clubs
- The diagram visually supports Monaco's reputation for producing players like **Mbappé, Bernardo Silva, Fabinho, and Tchouaméni**.

5. structural insight

- The diagram reveals a **hub-and-spoke talent system**:
  - **Source countries** -> **high-value positions** -> **clubs with distinct recruitment strategies**
- Talent accumulation is **not evenly distributed**:
  - A few positions and clubs dominate player flows
  - Smaller clubs play a critical intermediary role in the global ecosystem

Overall, this chord diagram shows a global football talent pipeline: a handful of high-output countries supply most players, those players are largely concentrated into high-demand positions (especially attackers and central roles like ST, CAM/CM/CDM, and CB), and then flow into a set of club hubs that recruit internationally. The heavy crossing ribbons indicate that recruitment is highly international rather than country-to-club exclusive, while differences in ribbon thickness suggest distinct club strategies-some clubs draw broadly across many nationalities and positions, whereas others show more focused profiles. In this network, AS Monaco stands out as a prominent receiving hub, connected to many nationalities and multiple positions, consistent with a club that attracts diverse talent (often younger profiles) and functions as a key intermediary in the wider transfer ecosystem.

Country Rankings by Average Overall Rating of Top-10 Players (2017–2022)

Description

This bump chart illustrates changes in **national elite football strength** from 2017 to 2022. For each year, countries are ranked based on the **average Overall rating of their top-10 players** (FIFA ratings). A **lower rank indicates a stronger elite talent pool**. The visualization highlights both **long-term stability** and **short-term fluctuations** in national football dominance.

Calculations for construction of bump chart

To construct the bump chart, the combined FIFA 2017–2022 player-level dataset was aggregated to the nationality–year level. For each nationality in each year, the total number of players was calculated, along with the maximum Overall rating and the average Overall rating of the top ten highest-rated players. The top-ten average was used as the primary performance metric to represent national team strength, as it emphasizes elite player quality while minimizing the influence of lower-rated players. Nationalities were then ranked within each year using a dense ranking method based on this metric, where higher averages received better ranks and ties shared the same rank. To maintain visual clarity, only the top 25 ranked nationalities per year were retained, producing a bump-ready dataset that enables clear comparison of changes in national team rankings over time.

- **Number of Players (n_players)**

This variable counts how many players belong to a given nationality in a specific year. It was calculated using a conditional counting formula that matches both year and nationality. COUNTIFS(Year_Column, fifa_Year, Nationality_Column, Nationality_Clean)

- **Average of Top 10 Overall Ratings (top10_overall)**

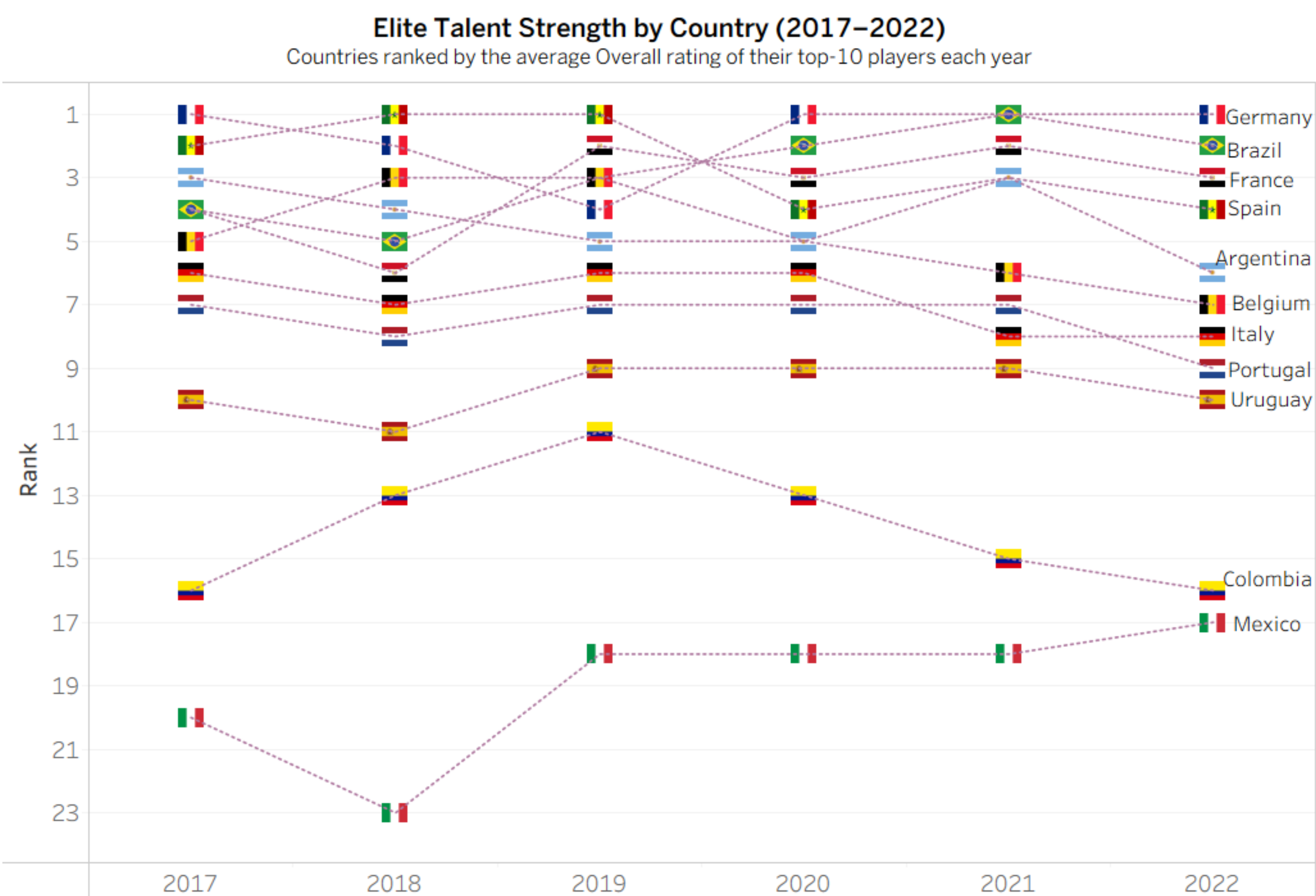
This variable is the primary metric used in the bump chart. It represents the average Overall rating of the top 10 highest-rated players from each nationality in each year. AVERAGE(LARGE(Overall_Ratings, {1,2,3,4,5,6,7,8,9,10}))

- **Maximum Overall Rating (max_overall)**

This variable captures the highest individual Overall rating within each nationality and year. It provides additional context on whether a nation's performance is driven by a single standout player. MAXIFS(Overall_Column, Year_Column, fifa_Year, Nationality_Column, Nationality_Clean)

- **Rank (rank)**

To support the bump chart visualization, each nationality was ranked within each year based on the top10_overall metric. A dense ranking method was used, where tied values receive the same rank and no ranks are skipped. 1 + COUNTIFS(Year_Column, fifa_Year, top10_overall_Column, ">" & top10_overall)



This bump chart tracks changes in national elite player strength from 2017 to 2022. For each year, countries are ranked by the average Overall rating of their top ten players. Lower ranks indicate stronger elite talent pools, highlighting both stability and shifts over time.

Key Insights

1. **Consistent Elite Nations at the Top**
   - **Germany, Brazil, France, and Spain** consistently occupy the **top 1–4 ranks** throughout the period.

- While minor rank swaps occur, these nations demonstrate **structural stability**, reflecting deep talent pools and sustained player development systems.

2. **Brazil and France Show Strong Resilience**
    - Brazil remains firmly within the **top two positions** in most years, underscoring the continued global impact of Brazilian elite players.
    - France shows a notable **upward trajectory post-2018**, aligning with its World Cup success and the emergence of younger elite stars.
3. **Mid-Tier Volatility**
    - **Argentina, Belgium, Italy, and Portugal** display noticeable **rank fluctuations**, moving between ranks 5–9.
    - These shifts suggest greater dependence on **a smaller set of star players**, where retirements, form changes, or generational transitions have a stronger impact on national averages.
4. **Declining or Lower-Tier Nations**
    - **Uruguay, Colombia, and Mexico** consistently rank in the **lower half**, with Colombia showing a gradual downward trend after 2019.
    - These countries show limited elite depth, where changes in a few players significantly affect national rankings.
5. **Stability vs. Mobility Contrast**
    - The chart clearly separates nations with **stable elite ecosystems** (top tier) from those with **high mobility and volatility** (mid and lower tiers).
    - Rank lines crossing frequently in the middle section emphasize competitive parity among mid-tier football nations.

Interpretation & Implications

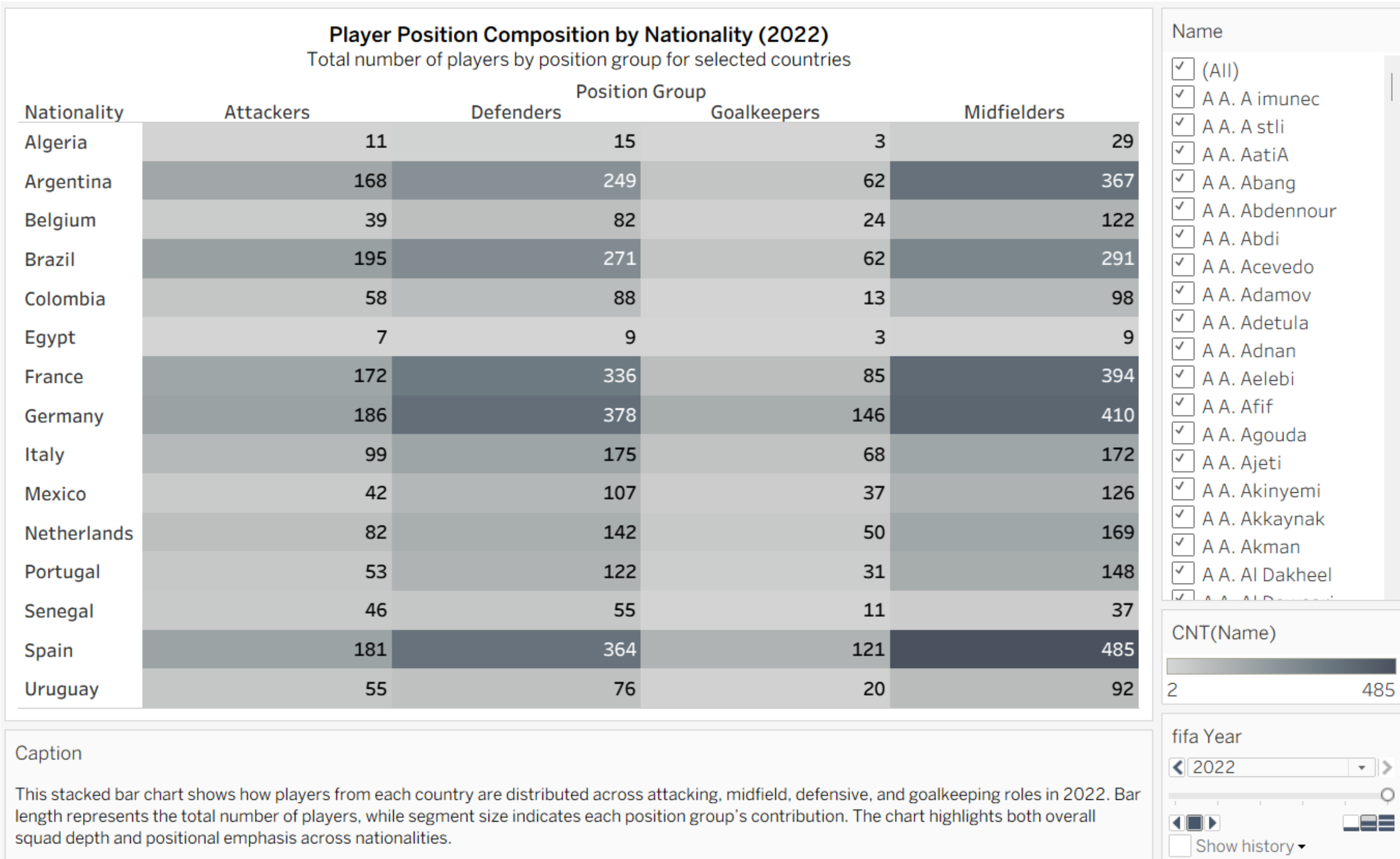
- **Elite football strength is path-dependent**: countries with established academies, league strength, and international exposure maintain dominance over time.
- **Rank volatility increases as average rating decreases**, indicating that weaker elite pools are more sensitive to individual player performance changes.
- The bump chart effectively captures **temporal dynamics**, making it easier to compare not just rankings, but **momentum and consistency** across nations.

Overall, the visualization demonstrates that while global football dominance remains concentrated among a small group of nations, **competitive reshuffling occurs primarily in the middle and lower ranks**. This highlights the importance of sustained talent pipelines over reliance on short-term star performance.

Player Position Composition by Nationality (2022)

Description

This heatmap presents the **distribution of football players by position group**—Attackers, Defenders, Midfielders, and Goalkeepers—across selected national teams for the year 2022. Each cell represents the **total number of players** from a country in a given position group, with darker shading indicating higher counts. The visualization enables comparison of **squad depth** and **positional emphasis** across nationalities.



Nationality	Attackers	Defenders	Goalkeepers	Midfielders
Algeria	11	15	3	29
Argentina	168	249	62	367
Belgium	39	82	24	122
Brazil	195	271	62	291
Colombia	58	88	13	98
Egypt	7	9	3	9
France	172	336	85	394
Germany	186	378	146	410
Italy	99	175	68	172
Mexico	42	107	37	126
Netherlands	82	142	50	169
Portugal	53	122	31	148
Senegal	46	55	11	37
Spain	181	364	121	485
Uruguay	55	76	20	92

Caption

This stacked bar chart shows how players from each country are distributed across attacking, midfield, defensive, and goalkeeping roles in 2022. Bar length represents the total number of players, while segment size indicates each position group's contribution. The chart highlights both overall squad depth and positional emphasis across nationalities.

Key Observations

1. **Midfield Dominance Across Elite Nations**
   - Midfielders constitute the **largest share** of players for most top footballing nations.
   - **Spain (485), Germany (410), and France (394)** show particularly strong midfield depth, reflecting tactical systems that emphasize ball control, passing, and transitional play.
2. **Defensive Strength in European Teams**
   - European nations such as **Germany (378), Spain (364), and France (336)** display high defender counts.
   - This suggests well-developed defensive pipelines and structured league systems that produce defenders at scale.
3. **Balanced Attacking Pools**
   - **Brazil (195), Germany (186), Spain (181), and France (172)** lead in attacking players, indicating sustained production of forward talent.
   - These nations maintain a balance between creativity in attack and overall squad depth.
4. **Goalkeepers as a Smaller, Specialized Group**
   - Goalkeepers represent the **smallest positional group** across all countries, which is expected due to role specialization.
   - Germany (146) and Spain (121) stand out, highlighting strong goalkeeper development programs.
5. **Smaller Footballing Nations Show Limited Depth**
   - Countries such as **Algeria, Egypt, Senegal, and Uruguay** show lower overall player counts across all positions.
   - These teams rely more heavily on a **narrower player base**, making positional shortages more impactful at the national level.

Comparative Insights

- **European powerhouses** exhibit both **high total player counts** and **balanced positional distributions**, indicating robust domestic leagues and youth systems.
- **South American nations** such as Brazil and Argentina display strong attacking and midfield numbers but relatively fewer goalkeepers, suggesting role specialization differences.
- **Midfield depth appears to be the clearest differentiator** between elite and mid-tier national team

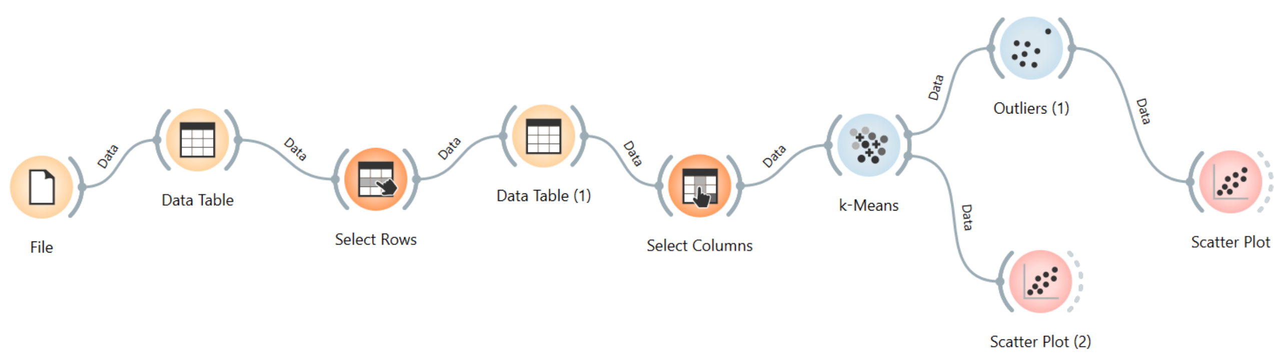
Interpretation & Implications
This visualization highlights how **positional depth reflects national football identity and development priorities**. Countries with strong midfield and defensive depth tend to perform consistently at the international level, as these positions contribute heavily to match control and tactical flexibility. Conversely, nations with limited positional depth face greater challenges in squad rotation and long-term competitiveness.

Overall, the heatmap reveals that **elite football nations are distinguished not only by star players but by deep, balanced positional pipelines**, particularly in midfield and defense. This positional composition provides structural resilience and supports sustained international success.

Advanced Analytics: Similarity, Dimension Reduction, and Outlier Detection (Orange Analytics)

To move beyond basic descriptive statistics, this project applies advanced analytical techniques implemented entirely in Orange Analytics to support exploratory and comparative analysis of professional football players. These methods focus on understanding player-level structure, similarity relationships, and anomalous performance profiles that are not immediately apparent from raw tabular data.

Rather than analyzing relational links between clubs, leagues, or nationalities, this stage of the analysis examines player attribute space, emphasizing how players compare to one another based on performance and development characteristics.



K-means Clustering of Player Performance Profiles

K-means clustering was applied to group players into distinct performance archetypes based on their technical and physical attributes. In the resulting visualization, each point represents a player positioned in the Overall–Potential feature space, while colors indicate cluster membership. This clustering reveals clear stratification within the player population, separating elite performers, high-potential players, and lower-rated profiles into distinct groups.

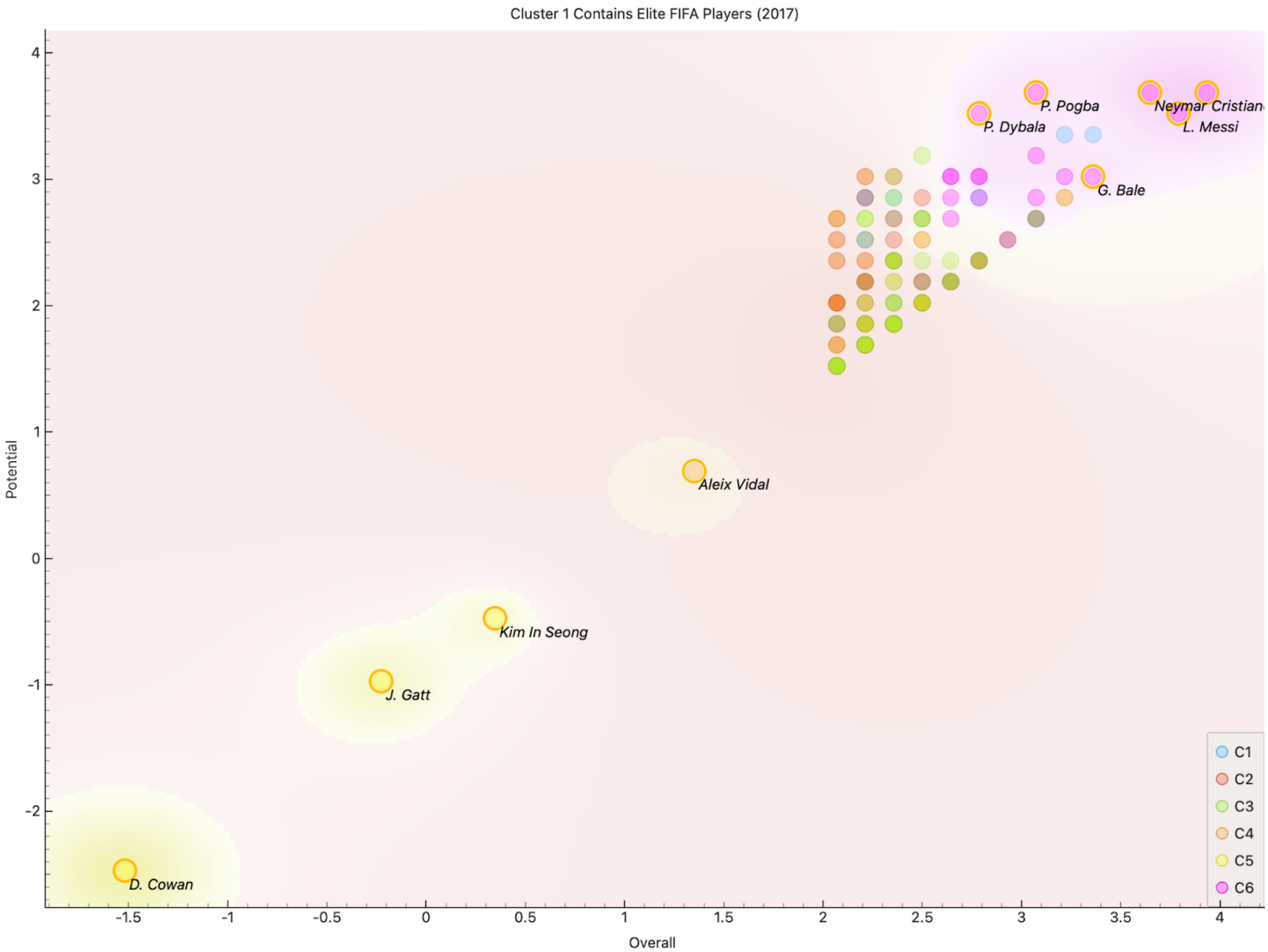


Figure: Local Outlier Factor (LOF) visualization in the Overall–Potential space, highlighting elite performers and structurally atypical player profiles.

One cluster in the upper-right region contains elite players such as **Lionel Messi**, **Cristiano Ronaldo**, and **Neymar Jr.**, characterized by both high current ability and high potential. Adjacent clusters capture strong but slightly less dominant players, including **Kylian Mbappe**, **Paulo Dybala**, and **Gareth Bale**, who exhibit high performance levels with more variation in potential. Lower clusters correspond to players with moderate or developing profiles, reflecting differences in skill maturity and physical attributes.

Overall, the K-means results highlight meaningful structure in the data by organizing players into interpretable performance tiers. This clustering provides a useful foundation for subsequent analyses, including silhouette evaluation to assess cluster quality and outlier detection methods to identify players whose profiles deviate substantially from their assigned group.

Outlier Detection Using Local Outlier Factor (LOF)

To identify players whose profiles deviate from typical development patterns, Local Outlier Factor (LOF) is applied within Orange Analytics. Local Outlier Factor (LOF) measures how isolated a player is relative to nearby players in the attribute space by evaluating local density. This approach highlights players with unusual combinations of overall ability and growth potential, allowing the detection of individuals who deviate from their nearest neighbors rather than from the dataset as a whole.

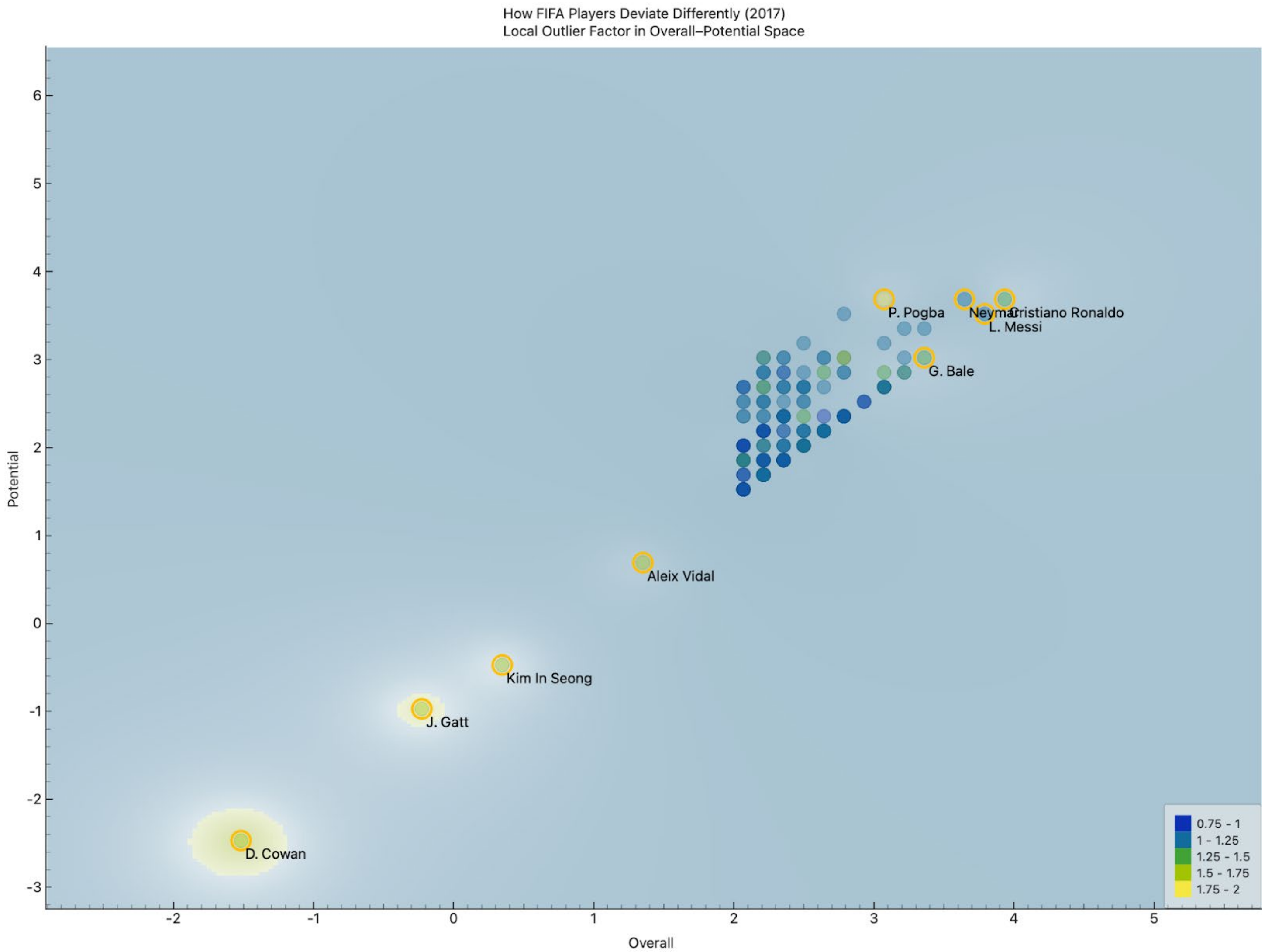


Figure. Local Outlier Factor (LOF) projection of players in overall–potential space using Orange Analytics. Outliers are identified based on LOF score -local deviations from neighboring player profiles**.**

In the visualization, most players form a dense cluster reflecting typical relationships between overall rating and potential. However, several players are highlighted as outliers due to their isolation or atypical positioning. **Elite players such as Lionel Messi and Cristiano Ronaldo** appear as local outliers at the upper extreme of the distribution, reflecting performance levels that significantly exceed those of their nearest peers. While they are not anomalous in a negative sense, their exceptional quality makes them statistically distinct within the dataset.

Additionally, players such as **Alexis Vidal**, **Kim In Seong**, and **D. Cowan** are identified as outliers in lower-density regions of the feature space. These players differ from the main population either due to unusually low ratings or uncommon combinations of overall performance and potential. The LOF color gradient emphasizes varying degrees of deviation, with darker highlights indicating stronger outlier behavior.

Overall, this analysis demonstrates that LOF effectively captures both **positive extremes (elite talent)** and **structural anomalies** in player profiles. These results complement the clustering analysis by identifying individuals whose attributes do not conform to typical group patterns, providing a nuanced understanding of exceptional and atypical players in the dataset.

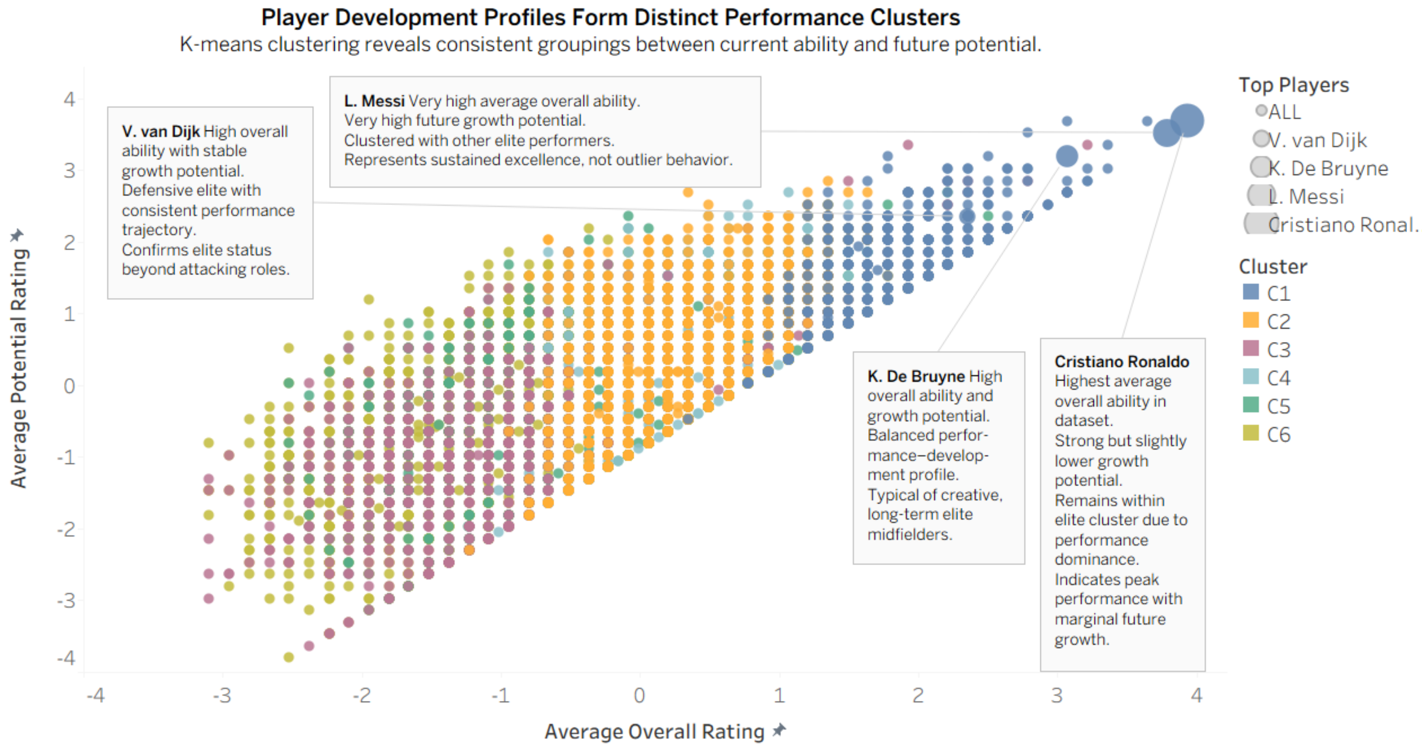
Analytical Insight

Python

After orange analytics, python code was generated for further analysis.

Python analytics first filters the combined FIFA dataset to include only players from the 2017 season and derived composite performance metrics (pace, shooting, passing, defending, and physicality) from individual attribute ratings. Local Outlier Factor (LOF) with Manhattan distance was applied to identify anomalous player profiles, producing an LOF score, rank, and categorical outlier buckets. Players were additionally grouped using K-means clustering, and silhouette scores were computed to assess cluster quality. The resulting dataset supports position-aware, cluster-based, and outlier-focused visual analysis in Tableau.

Cluster Analysis: Player Development Profiles (Overall vs Potential)



K-means clustering separates players into consistent groups based on the relationship between current overall ability and future potential. Elite players such as Messi, De Bruyne, and Ronaldo occupy the high-ability cluster, while lower-rated clusters reflect differing development paths rather than random variation.

This chart visualizes player clusters based on their overall rating and potential, revealing structural groupings in player development profiles across the FIFA dataset.

Elite cluster: high overall and high potential

A dense cluster emerges in the upper-right region of the plot, representing elite players with both high current ability and strong future potential.
Players such as Lionel Messi, Cristiano Ronaldo, Kevin De Bruyne, Mohamed Salah, and Virgil van Dijk all appear within this cluster, despite differences in position and playing style.
This concentration indicates that elite status is characterized by structural similarity in development profiles rather than isolated individual excellence.

Mid-tier clusters: balanced and developing profiles

Several intermediate clusters occupy the central region of the plot, capturing players with moderate overall ratings and varying potential ceilings.
These clusters reflect players in developmental or transitional stages, including emerging talents and established professionals who are no longer expected to improve significantly.
The spread within these clusters highlights heterogeneity in growth trajectories among non-elite players.

Lower clusters: limited growth ceilings

Clusters located toward the lower-left region correspond to players with lower overall ability and limited potential. These players tend to follow constrained development paths, with minimal upward mobility relative to the rest of the dataset.

Strong positive relationship between overall and potential

Across all clusters, a clear positive relationship between overall rating and potential is observed. Players generally progress along a diagonal trajectory, reinforcing that future potential is closely tied to current performance levels at the population scale.

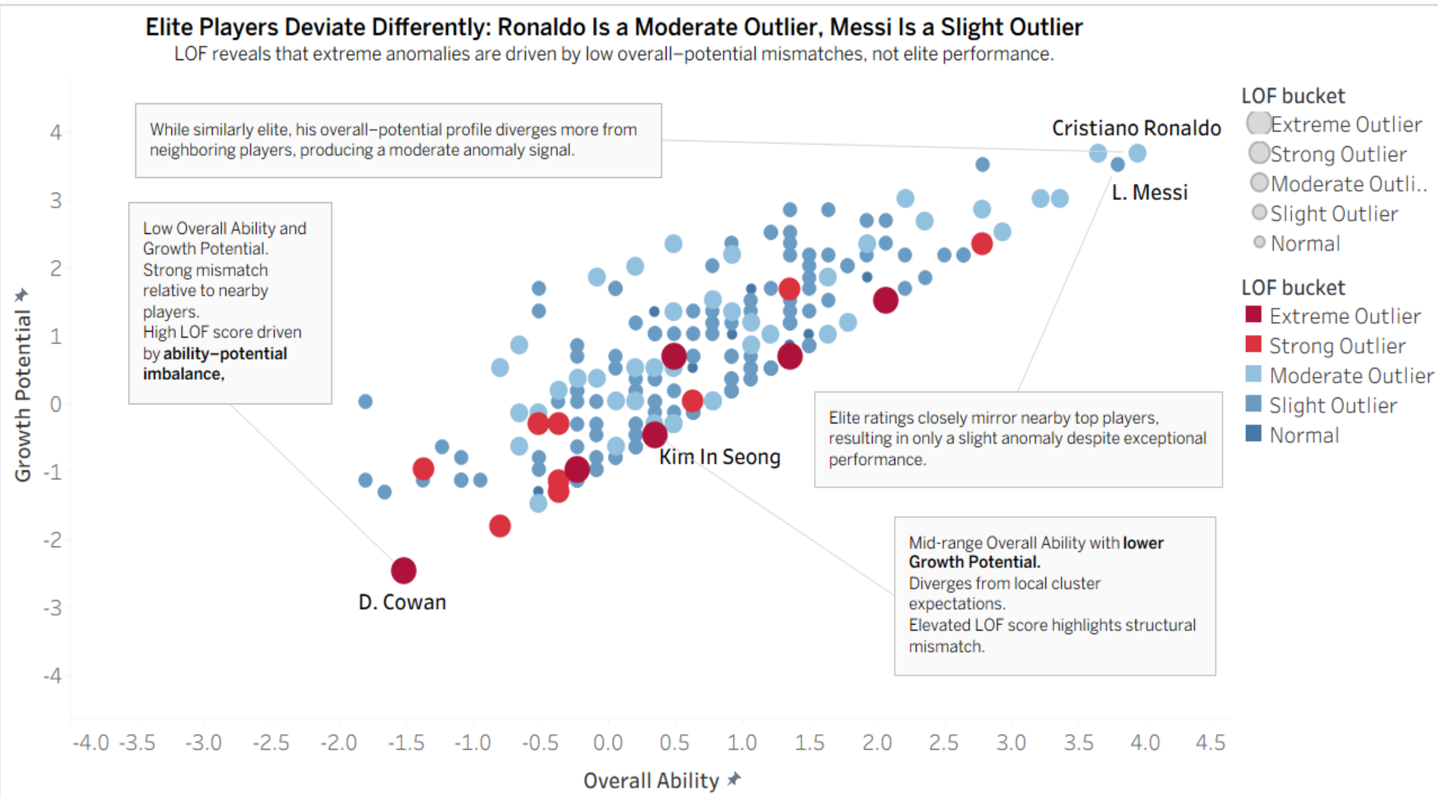
Takeaway

Despite covering a wide range of player types, the cluster structure reveals a highly ordered development landscape:

- Elite players form a compact, high-performance cluster
- Mid-tier players exhibit diverse but structured growth patterns
- Lower-tier players face constrained development ceilings

Outlier Analysis: Local Outlier Factor (LOF) in Overall–Potential Space

This chart presents Local Outlier Factor (LOF) results applied to overall–potential space, identifying players whose development profiles deviate from nearby peers.



While most players cluster along a strong positive relationship between overall ability and growth potential, Local Outlier Factor (LOF) highlights players whose profiles diverge from nearby peers. Cristiano Ronaldo appears as a moderate outlier due to a higher overall–potential mismatch, whereas Lionel Messi remains only slightly anomalous, reflecting exceptional performance that closely mirrors other elite players rather than statistical isolation.

Outliers concentrated at the edges of the distribution

LOF-identified outliers are primarily located along the boundaries of the overall–potential distribution rather than at its center. These anomalies often reflect players with mismatched overall ability and potential, such as relatively low-rated players with unusually high potential or vice versa.

Elite players are not extreme anomalies

Contrary to intuition, elite players do not consistently appear as strong outliers.
Lionel Messi exhibits only a slight anomaly signal, as his ratings closely align with other top-tier players in the elite cluster.
Cristiano Ronaldo shows a moderate anomaly score, reflecting greater divergence from nearby peers, but still remains structurally close to the elite population.

Strong anomalies driven by developmental mismatch

The strongest LOF scores are observed among lower-rated players whose potential ratings differ sharply from surrounding players. These cases represent atypical development paths rather than exceptional performance, highlighting that LOF captures local inconsistency rather than global extremity.

LOF complements clustering, not contradicts it

While clustering reveals broad structural groupings, LOF isolates individual players who break local patterns within those groups. Together, these methods show that anomaly detection and similarity analysis capture distinct but complementary aspects of player development.

Takeaway
LOF analysis confirms that:

- Anomalies are driven by overall–potential mismatch, not elite performance
- Elite players cluster tightly and rarely appear as extreme outliers
- Developmental inconsistency is more common among lower- and mid-tier players

---